GSLC 1 Deep Learning

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1. **Definition of the task**

For the deep learning project, we will be creating a semantic image segmentation model, based on the U-Net Convolutional Networks for Biomedical image segmentation in PyTorch.

2. **The Input and Output of the model**

The Input for the model is a collection of Brain Tumor MRI Images containing 2146 images that's separated into train, test, and validation dataset each containing 1502, 429, and 215 images. Every image will be 640 x 640 pixels in resolution. The output of the model is the classification of every pixel in the images, whether it's class 1 (non-tumor) or class 2 (Tumor).

3. **Metric to Evaluate Model**

The metric for the model will be Jaccard index, we're using Jaccard index because of its use to understand the similarities between sample sets/mask, meaning we can evaluate the model based on the similarities in the mask of the original sample and the predicted sample with the most optimal being 1 (Completely the same). For this dataset, if the pixel containing the tumor on the original sample and the predicted sample is completely the same, the Jaccard index will be 1. With it decreasing for each miss classification.

4. **Introduction of the topic**

Image Segmentation of Brain tumor MRI images. This image segmentation will be done by masking the images into 2 classes. With the area having pixel with value of 0 as normal cells (Non-Tumor) and value of 1 being tumor cell (Tumor). By segmentizing the areas that's predicted to be tumor, doctors can make an analysis of the MRI images more easily with minimal consultation to radiologist before or after hand, this segmentation will make doctor more easily explain to patient on where the tumor is too.

5. **State the dataset you have chosen.**

The dataset that we have chosen is the Brain Tumor Image Dataset: Semantic Segmentation that's designed specifically for semantic image segmentation that's sourced from Kaggle.

Brain Tumor Image DataSet : Semantic Segmentation (kaggle.com)